

Evaluation Quiz

Deep Learning: A mathematical physics approach
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Instructions

Select the correct answer for each of the following 20 multiple-choice questions, which are based entirely on the concepts covered in *Lecture 1*.

Questions

Question 1. What is the main challenge currently related to the terabytes of data that computers capture and store every day?

- A) Erasing them safely and efficiently.
- B) Doing something useful with this data.
- C) Compressing them to save server space.
- D) Transforming them into analog formats.

Question 2. According to the presentation, what is the key concept of "Learning"?

- A) Memorizing large volumes of linear text.
- B) Learning from data by adjusting and adapting to new circumstances through experience.
- C) Programming explicit instructions into a computer.
- D) Performing mathematical calculations at high speed.

Question 3. Which of the following actions is **not** mentioned as an important aspect of animal learning?

- A) Remembering.
- B) Programming.
- C) Adapting.
- D) Generalizing.

Question 4. How is Artificial Intelligence (AI) defined?

- A) The overarching field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence.
- B) Mathematical algorithms dedicated exclusively to recognizing images in high dimensions.
- C) A system of integrated circuits that physically replaces humans.
- D) Linear programming of structured data.

Question 5. What primarily distinguishes Machine Learning (ML) from traditional programming?

- A) It requires quantum processors to function.
- B) It is explicitly programmed for every specific task.
- C) Instead of being explicitly programmed, it identifies patterns within a dataset to make predictions.
- D) It is only used in statistical text analysis.

Question 6. Historically, in the development of Machine/Deep Learning, which period is associated with "Connectionism"?

- A) 1940 - 1960
- B) 1980 - 1990
- C) 2006 - Present
- D) 1900 - 1920

Question 7. Among the types of machine learning, which one is characterized by learning from examples using targets or labels?

- A) Unsupervised Learning
- B) Reinforcement Learning
- C) Evolutionary Learning
- D) Supervised Learning

Question 8. In the context of Machine Learning, what is the central goal of Regression?

- A) Deciding which of N classes a data point belongs to.

- B) Fitting a mathematical function describing a curve so that the curve passes as close as possible to all data points.
- C) Finding anomalous or outlier values in a set of vectors.
- D) Artificially reducing the dimensionality of a spatial dataset.

Question 9. What restriction or characteristic is fundamental and important when talking about Classification models?

- A) It is discrete!
- B) It requires quantum wave functions.
- C) It only works with two independent variables.
- D) It always results in continuous and infinite values.

Question 10. What does the "Curse of dimensionality" refer to?

- A) That mathematics in higher dimensions are impossible to compute.
- B) That neural networks collapse if they have more than three dimensions.
- C) If the number of input dimensions grows, the number of data points required increases faster.
- D) That high dimensions force the exclusive use of Unsupervised Learning.

Question 11. According to "The Machine Learning Process", what is the correct order of the first three steps?

- A) Data collection → Training → Evaluation
- B) Algorithm choice → Feature selection → Training
- C) Data collection → Feature selection → Algorithm choice
- D) Parameters selection → Evaluation → Training

Question 12. In the biological study of learning in the brain, what does "Neuroplasticity" describe?

- A) The strength of synaptic connections modifies or new connections are created.
- B) The physical size of neurons is irreversibly reduced by half.
- C) A permanent refractory period is generated.
- D) The brain rejects input data due to excess dimensions.

Question 13. What does "Hebb's Rule" postulate about cellular learning?

- A) If two neurons consistently fire out of sync, their connection becomes immune to change.
- B) Neurons cannot modify their internal cellular structure.
- C) Pavlov's classical conditioning has no biological basis.
- D) If two neurons consistently fire simultaneously, then any connection between them will change in strength, becoming stronger.

Question 14. In the mathematical analogy of a neuron (perceptron), which element fulfills the biological function of the cell membrane that collects electrical charge?

- A) Weighted inputs.
- B) Adder.
- C) Activation function.
- D) Linear regression.

Question 15. With which biological specimen did Hodgkin and Huxley conduct their studies on neurons in 1952 to avoid the problems of human microscopic size?

- A) Giant squid.
- B) Chimpanzees.
- C) Albino mice.
- D) Dogs.

Question 16. From the perspective of Linear Algebra, what **additional** axiom must a group satisfy (along with closure, associativity, identity element, and inverse) to be considered an *Abelian Group*?

- A) The existence of a transposed matrix.
- B) The scalar distributive property.
- C) The commutative property.
- D) Linearity in the first argument.

Question 17. According to the "Theorem (Existence and Uniqueness of Coordinates)", any n -dimensional vector space over a field $\mathbb{F}$ is isomorphic to:

- A) The null space $\ker(T)$.

- B) $\mathbb{F}^n$.
- C) A multivariate affine transformation.
- D) An identity matrix of $m \times n$.

Question 18. What are the two conditions that a subset $\mathcal{B}$ must satisfy to be considered a "Basis" of a vector space V ?

- A) Being orthonormal and having unit norm.
- B) Being an infinite and symmetric set.
- C) Linear Independence and the Spanning Property.
- D) Having real eigenvalues and not being negative definite.

Question 19. According to the linear algebra definitions presented, what is a Linear Functional?

- A) A simple linear regression without parameters.
- B) A special case of a linear transformation where the codomain is the field of scalars $\mathbb{F}$.
- C) A three-dimensional orthogonal vector.
- D) The set of all possible bases of an infinite vector space.

Question 20. Integrating all the mathematical concepts presented, how can a single simple neuron be analytically conceptualized?

- A) As a purely unsupervised iterative classification.
- B) Exclusively as a static linear functional without a spatial limit.
- C) As a model analogous to a linear regression (multivariate) followed by an activation function.
- D) As the iterative and inverse process of the rank-nullity theorem.

Answer Key

Correct Answers

- Question 1: B
- Question 2: B
- Question 3: B
- Question 4: A
- Question 5: C
- Question 6: B
- Question 7: D
- Question 8: B
- Question 9: A
- Question 10: C
- Question 11: C
- Question 12: A
- Question 13: D
- Question 14: B
- Question 15: A
- Question 16: C
- Question 17: B
- Question 18: C
- Question 19: B
- Question 20: C